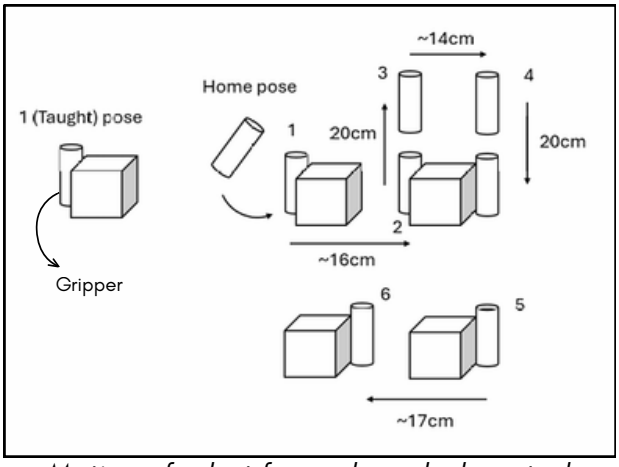
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UR5e AUTONOMOUS MANIPULATION TASKS- JHU 2025



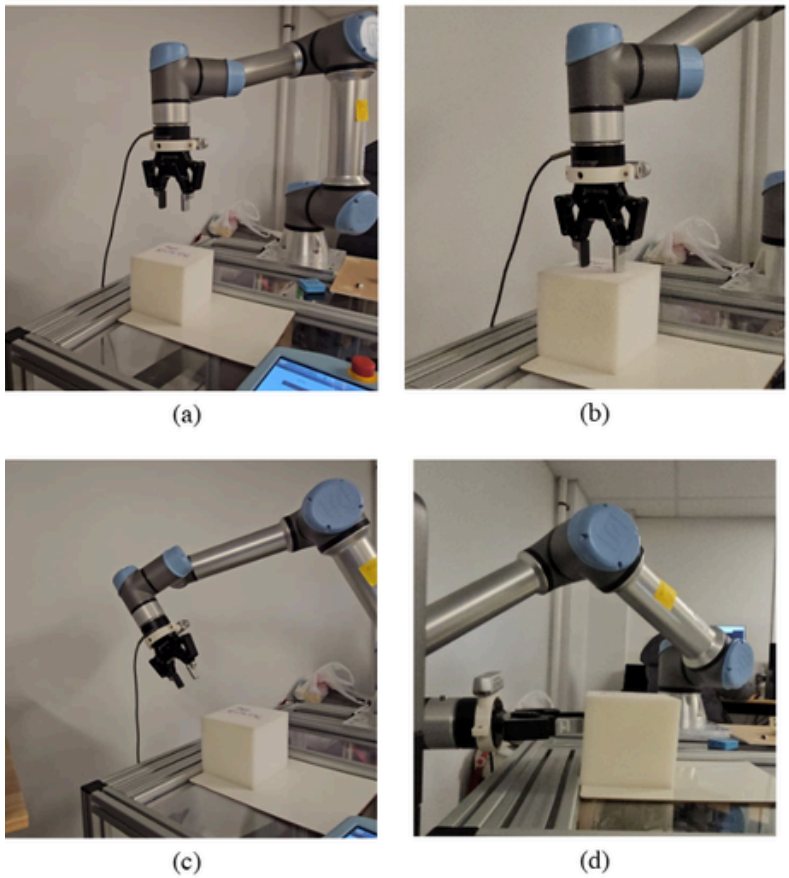
Motion of robot for push-and-place task

What?

- Used a UR5e manipulator to perform two tasks:
  - **Autonomous push-and-place task**: pushed a foam cube along a straight line, then autonomously executed a return push.
  - **Precision evaluation task**: assessed near-contact motion by approaching multiple surfaces of a foam cube without visible deformation.
- Implemented and compared Inverse Kinematics (IK) and Resolved-Rate (RR) Cartesian control for both.

How?

- Taught initial pose using freedrive mode and planned motion in SE(3).
- Implemented controllers in:
  - ROS 2 (IK + RR control, simulation and hardware)
  - MATLAB RTDE (RR control)
- Designed IK solution selection to:
  - Filter infeasible joint solutions
  - Maintain motion continuity
  - Avoid large joint jumps using joint-space proximity thresholds
- Developed control and execution code in MATLAB and Python.
- Validated performance in RViz simulation (ROS 2) and on real hardware



Motion of robot for precision evaluation task

Results

- Achieved accurate straight-line Cartesian pushing and consistent, repeatable near-contact positioning on the real robot.
- IK control produced smoother, faster motion with minimal tuning
- RR control achieved similar accuracy but required careful gain tuning
- ROS execution showed lower latency and better reliability than RTDE
- Validated the robustness of the IK-based control strategy for precision manipulation tasks
- **Watch both tasks in action below:**
  - Push-and-Place Task
  - Precision Evaluation Task

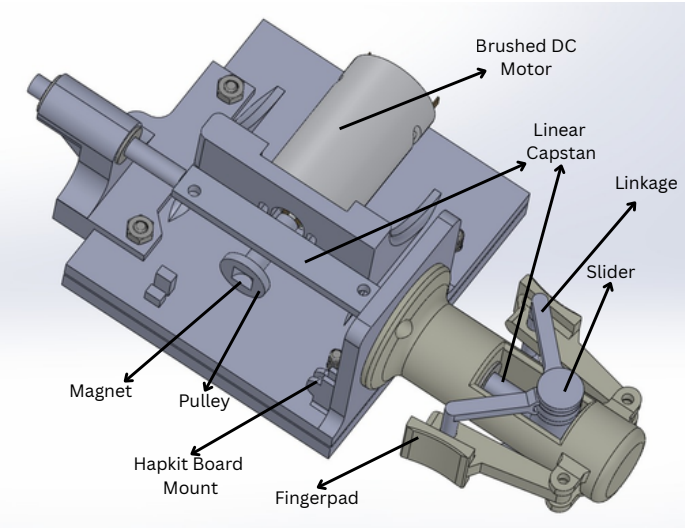
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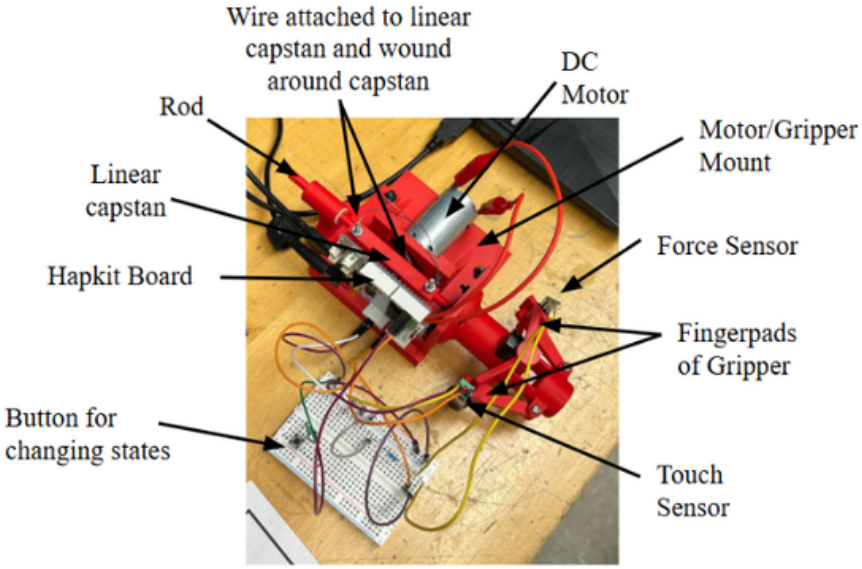
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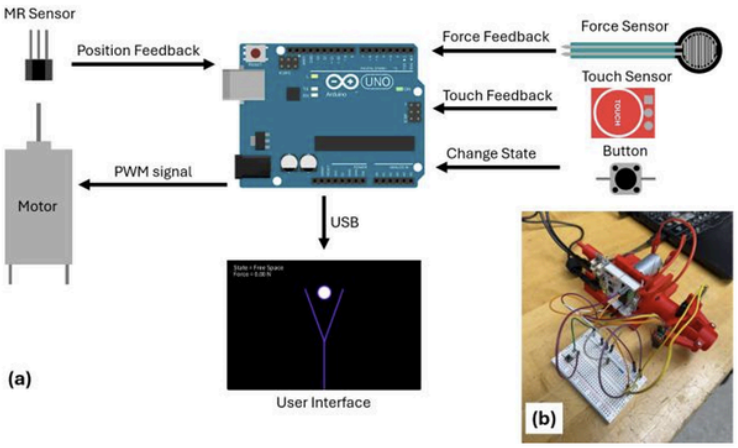
GRASPING FORCE FEEDBACK FOR DA VINCI SURGICAL ROBOTS - JHU 2025





What?

- Developed a kinesthetic haptic interface to add grasping force feedback to a da Vinci-style surgical gripper
- Addressed the lack of haptic feedback in robotic minimally invasive surgery (MIS)
- Designed to preserve the original ergonomics and workflow of the da Vinci leader hand
- Enables users to distinguish between:
  - Free space
  - Soft tissue
  - Hard tissue
- Goal: Provide intuitive perception of tissue stiffness without redesigning the surgeon interface



How?

- Designed and 3D printed a 1:1.2 scale da Vinci gripper replica
- Built a capstan-driven actuation system powered by a DC motor for force rendering
- Integrated MR position sensing + FSR-based force sensing for closed-loop interaction
- Implemented embedded control using the Hapkit 3.0 board (motor driver + sensing interface)
- Developed real-time C++ control architecture for stable haptic feedback
- Implemented physically-based models (spring, damping, friction, impact) for tissue simulation
- Applied signal filtering and state estimation for smooth, noise-robust force output
- Synchronized hardware with a real-time Processing simulation for visual–haptic coupling
- Enforced mechanical and software safety constraints (≤ 5 N force limit)

Results

- Successfully rendered distinct interaction states: free space, soft tissue, and hard tissue
- Achieved realistic grasping force ranges:
  - Free space: ~0 N
  - Soft tissue: 0–2 N
  - Hard tissue: 1–4 N
- Demonstrated smooth mechanical performance of the fully 3D-printed capstan-linkage system
- Identified motor commutation granularity (brushed DC motor) as the primary source of residual force ripple
- Despite time constraints, successfully delivered a fully functional working prototype (see demo video)
- Established clear path for industry-level smoothness via brushless motor + higher-resolution actuation
- **Watch the gripper in action here**

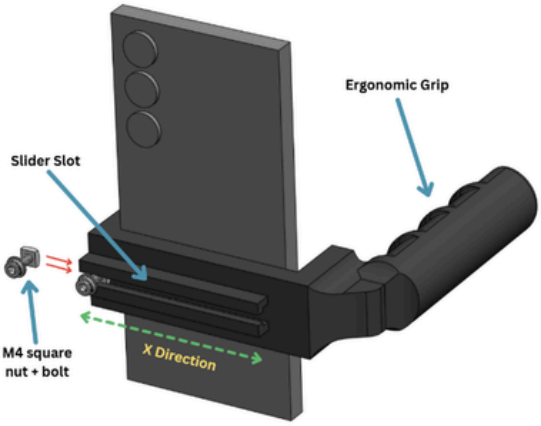
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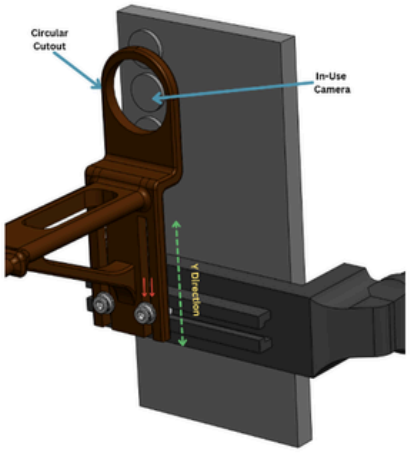
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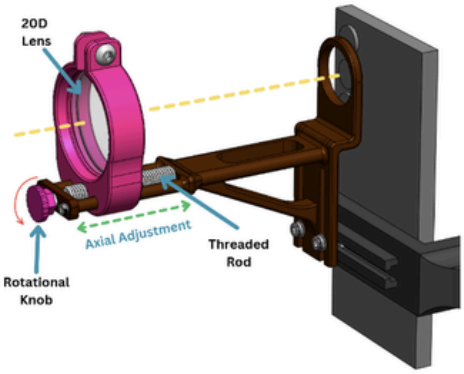
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CLARIFEYE: AFFORDABLE AI-BASED FUNDUS IMAGING - JHU 2025

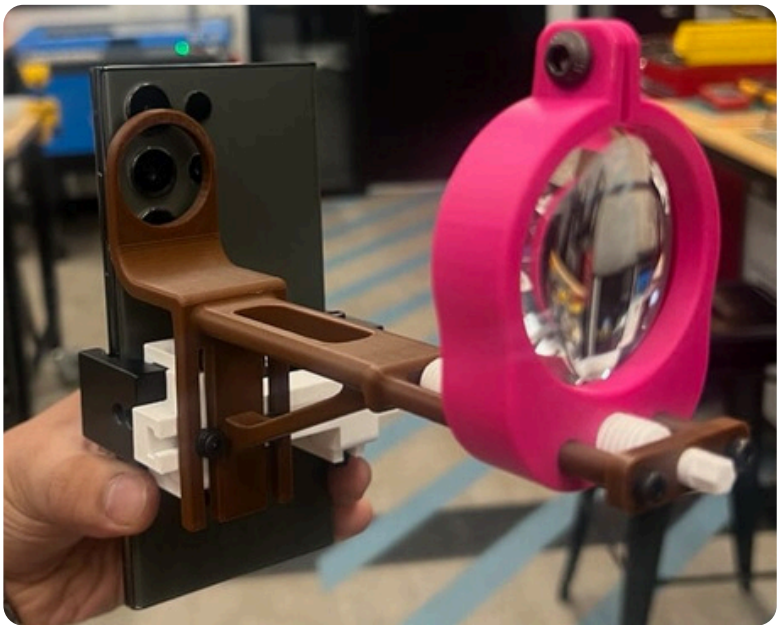






What?

- Co-founded Clarifeye, a startup providing scalable AI-powered retinal screening in LMICs.
- Led hardware design of a handheld, smartphone-agnostic fundus imaging attachment.
- Designed a modular, fully mechanical system integrating a 20D condenser lens, X–Y alignment, and axial lens adjustment.
- Ensured universal smartphone compatibility without device-specific adapters.
- Optimized hardware for low-cost (<$100) and rapid field deployment (<4 min screening).
- Hardware supports AI pipeline: image enhancement, stitching, GAN-based restoration, and disease classification.



How?

- Designed full CAD architecture in SolidWorks with modular components for easy 3D printing and replacement.
- Engineered X–Y planar alignment and axial lens adjustment to maintain concentric optical alignment across phones.
- Integrated a threaded Z-axis mechanism with guide rails and manual knob for precise focusing.
- Used lightweight PLA and standard fasteners for rapid prototyping and low cost.
- Developed ergonomic grip and rigid locking mechanisms to reduce motion artifacts.
- Iterated prototype on commercial mounts to refine alignment, stability, and usability.
- Reviewed patents and competitor designs to ensure universality, low cost, and manufacturability.

Results

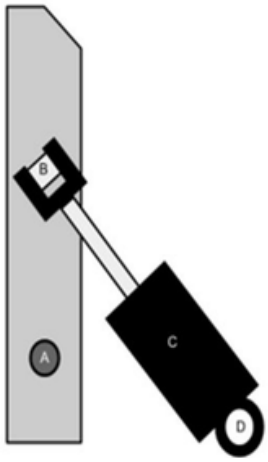
- Produced a functional handheld prototype with smartphone-agnostic alignment and axial focus control.
- Reduced hardware complexity compared to existing adapters and FDA-cleared devices.
- Achieved lightweight, ergonomic, modular design for CHW-friendly operation.
- Designed system to maximize image gradability for downstream AI processing.
- Supports multi-disease detection pipeline while remaining cost-effective.
- Positioned Clarifeye as a scalable, low-cost alternative to competitors.
- Contributed to market research, patent differentiation, and startup strategy.

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ADAPTABLE PEDAL BOX DESIGN - UIC 2024







What?

- Design of a pedal box for the Formula SAE team at UIC that is compatible for combustion and electric racecars

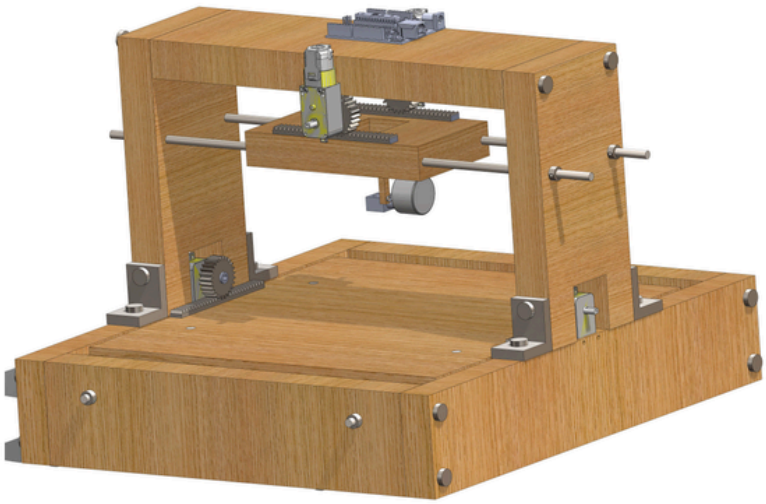
How?

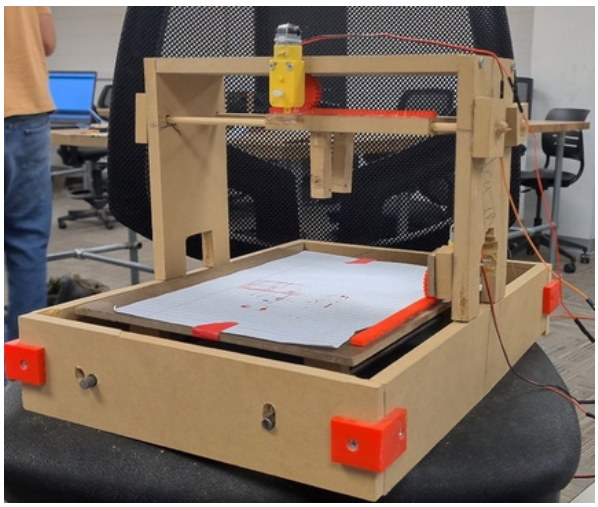
- Used SolidWorks to design all the iterations
- Simulated several iterations using ANSYS
- Created prototypes using laser-cut acrylic sheets and 3D printed parts

Results

- The final prototype fit perfectly into its designated area and adhered to all the FSAE rules
- Presented this project at EXPO UIC 2024 and won the "Best in show award"

AUTONOMOUS WRITING DEVICE - UIC 2021





What?

- Created a handmade autonomous writing device from MDF wood, 3D-printed custom parts, and an Arduino UNO microcontroller.
- Developed as a project for an Introduction to Design and Graphics course at UIC.

How?

- Designed and refined iterations in SolidWorks.
- Cut and shaped MDF wood sheets for structural parts using machine shop equipment.
- Integrated a 3D-printed rack-and-pinion mechanism for precise control.
- Programmed the device in C++ on an Arduino UNO microcontroller to enable automated functionality.

Results

- The final device successfully drew shapes and letters with stability and operated seamlessly at the push of a button.
- Presented the project to UIC professors, receiving valuable feedback on optimizing the design for reduced mass and cost.

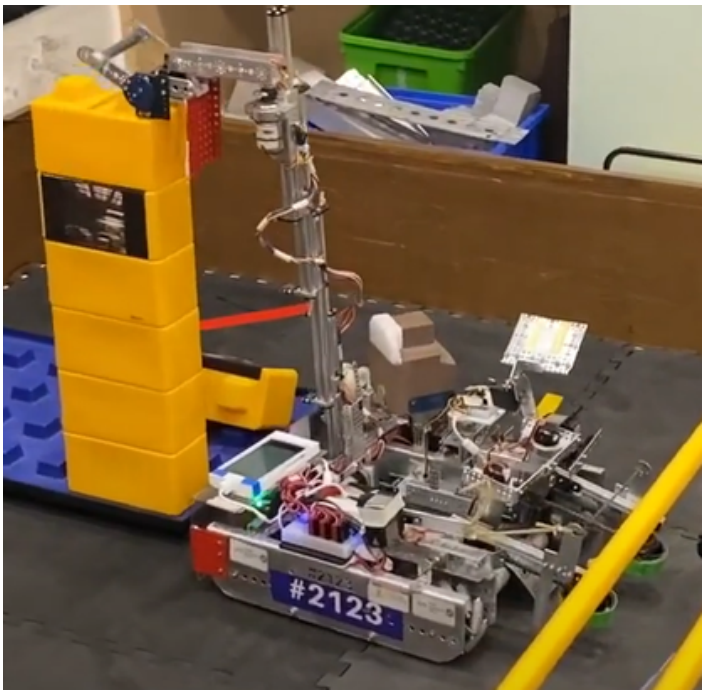
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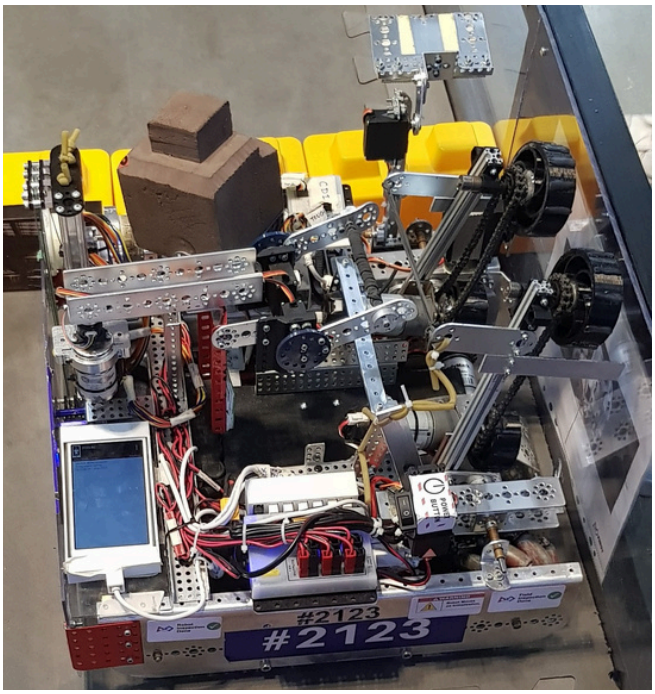
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FIRST TECH CHALLENGE (FTC) SKYSTONE 2019-20





What?

- Designed and built a robot for the FIRST Tech Challenge Skystone 2019-2020 competition in India as part of the team "Mad Engineers."
- The robot autonomously and manually collects and stacks large blocks on a tray within a 2.5-minute timeframe on the playing field.

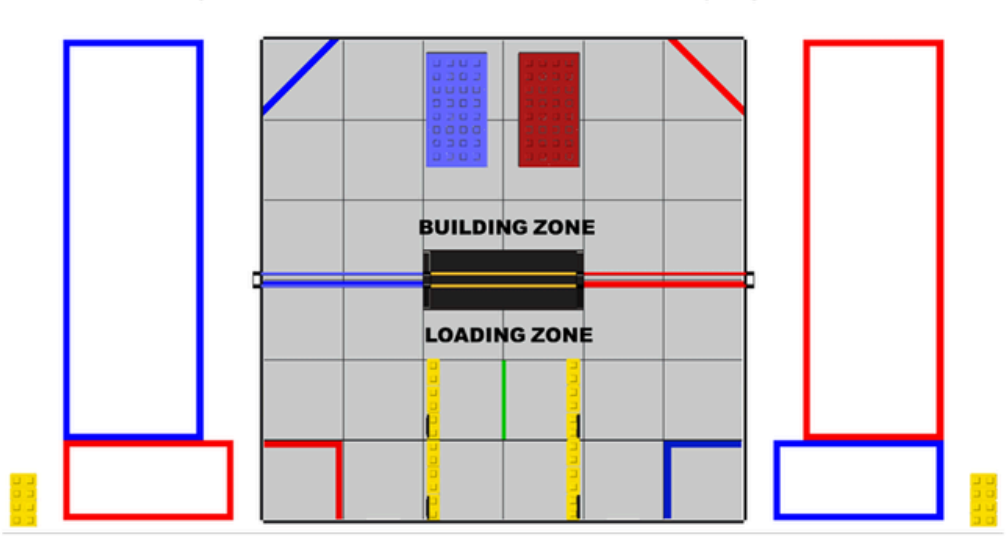
How?

- Designed using SolidWorks and iteratively prototyped and tested.
- Constructed with laser-cut metal, 3D-printed parts, and Tetrix components.
- Programmed in Java to enable both autonomous and manual wireless operation.

Results

- Awarded the Collins Aerospace Innovative Award for the most creative and innovative robot design.
- **View the robot completing tasks in the playing field here**

Top view of the playing field



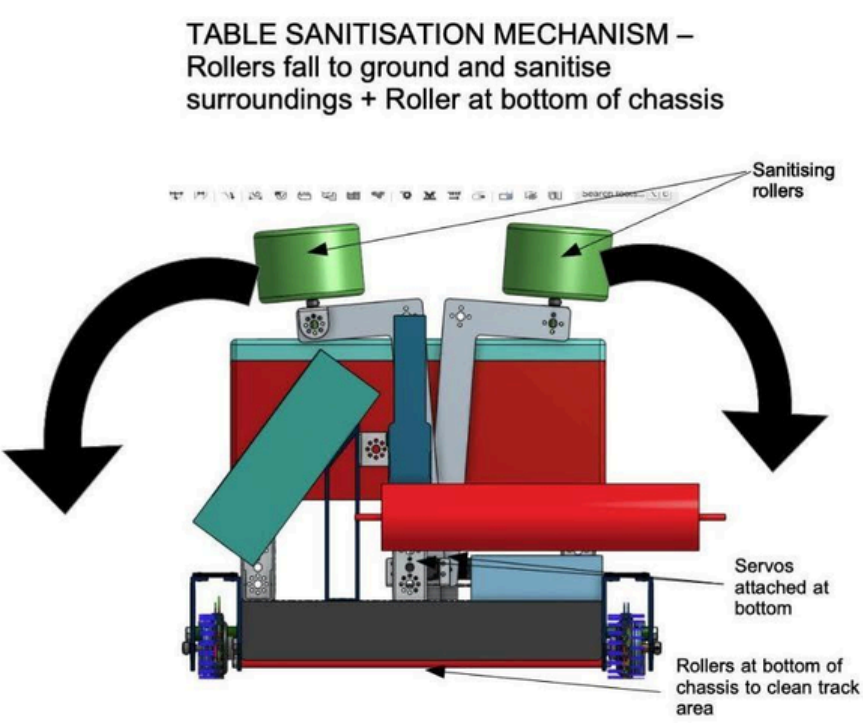
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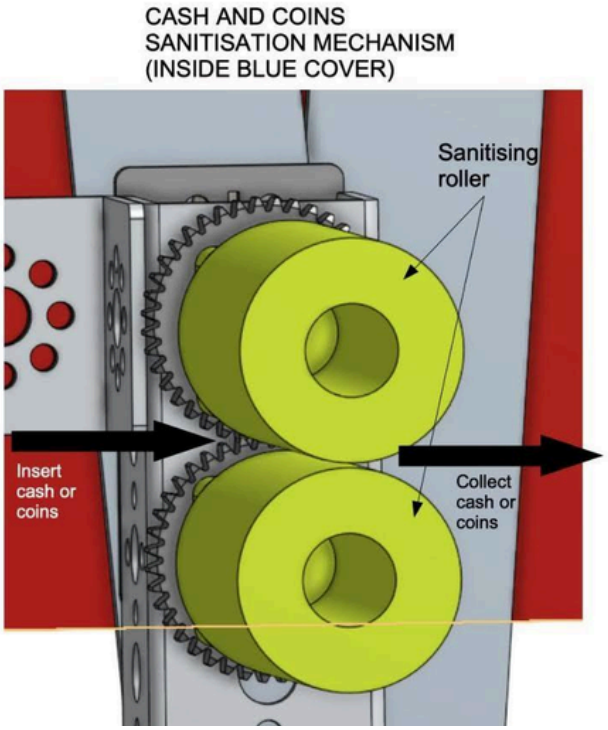
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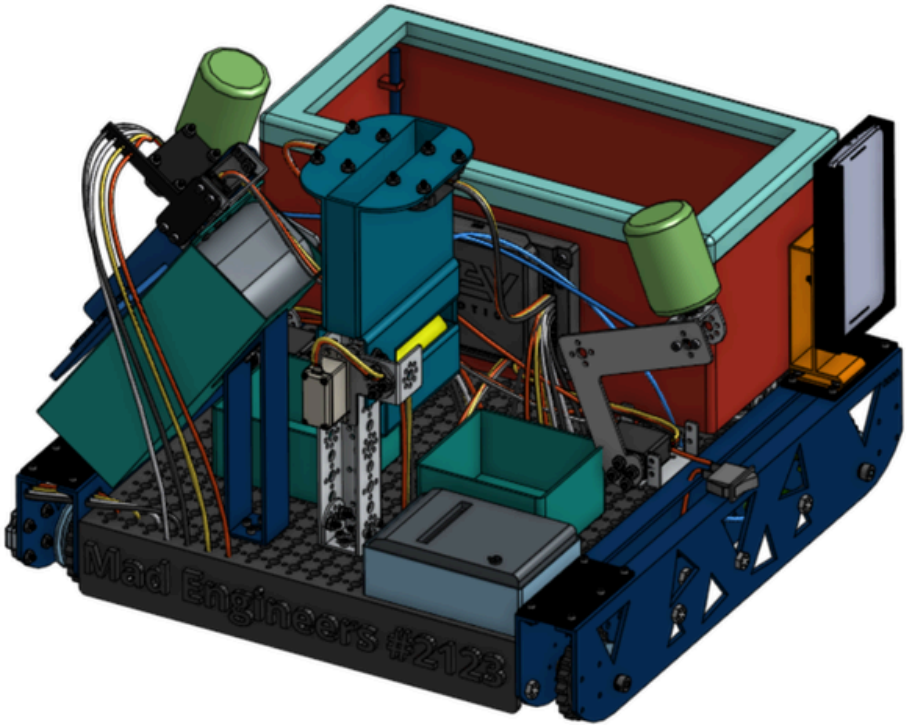
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ROBOTS TO THE RESCUE - FIRST® 2020







What?

- Designed a robot to replace the role of pharmacists, cashiers and security guards for Onshape's Robots to the Rescue competition for FIRST® in 2020.
- Automated the packaging and delivery process for supplies, incorporating a disinfection mechanism that sanitizes currency and the robot's workspace to ensure a contact-free, germ-free experience during transactions.

How?

- Developed detailed CAD models on Onshape for the robot, ensuring accurate and functional designs for each component.
- Worked closely with a diverse team to ensure the CAD designs aligned with both the technical and functional requirements of the project.

Results

- Our team ranked 18th out of 151 particpating teams
- The robot's design was well-received for its practicality in real-world applications, showcasing innovative solutions for contactless operations in essential services.

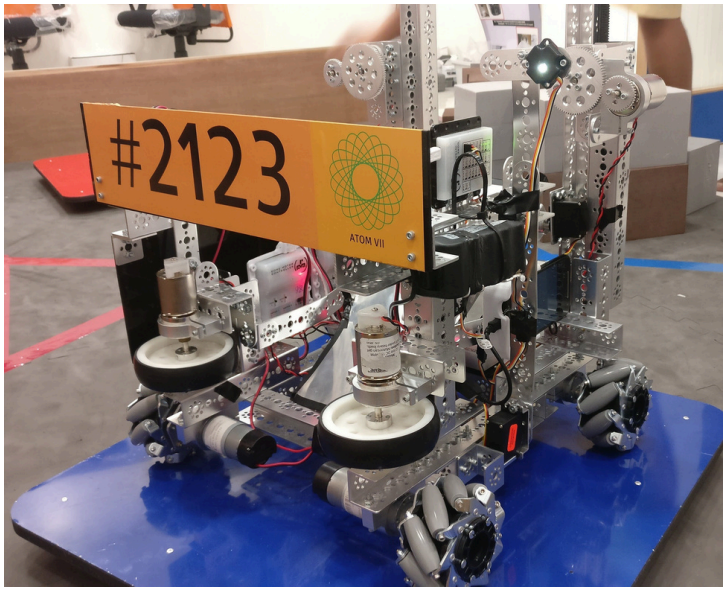
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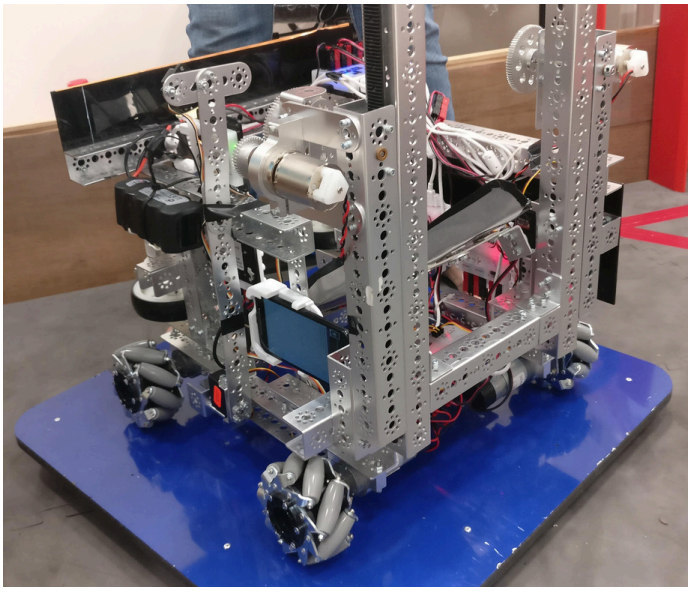
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FIRST TECH CHALLENGE (FTC) RELIC RECOVERY 2017-18





What?

- Designed and built a robot for the FIRST Tech Challenge Relic Recovery 2017-2018 competition in India as part of the team "Mad Engineers."
- The robot autonomously and manually collects and stacks 6-inch foam cubes into four columns within a 2.5-minute timeframe.

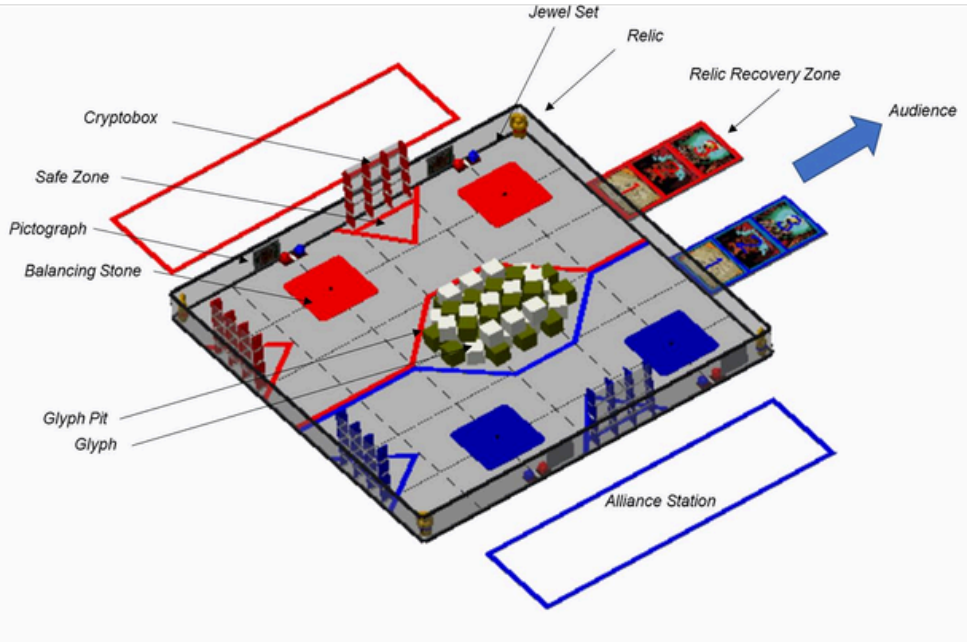
How?

- Designed using SolidWorks with seven iterations made and tested.
- Constructed with 3D-printed parts and Tetrix components.
- Programmed in Java for both autonomous and wireless manual control.

Results

- Awarded the Finalist Alliance 1st Runner Up award for securing second place in the overall competition.
- **View the robot competing in the semifinals here**

View of the playing field



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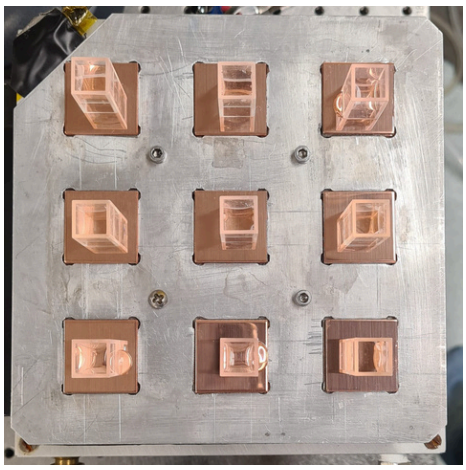
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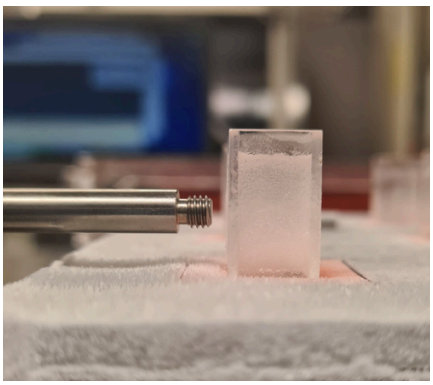
ICE ADHESION AND PHASE CHANGE MATERIAL RESEARCH - UIC 2022

What?

- Contributed to experimental research on ice adhesion strength using salt solutions on metallic surfaces.
- Conducted material characterization of polymers for cold-climate applications.
- Assisted in research on phase change materials (PCMs) for condensation and frost control.
- Credited for general assistance in the peer-reviewed publication: "How to Select Phase Change Materials for Tuning Condensation and Frosting," Advanced Functional Materials (Chatterjee et al., 2022). Click here to check out publication



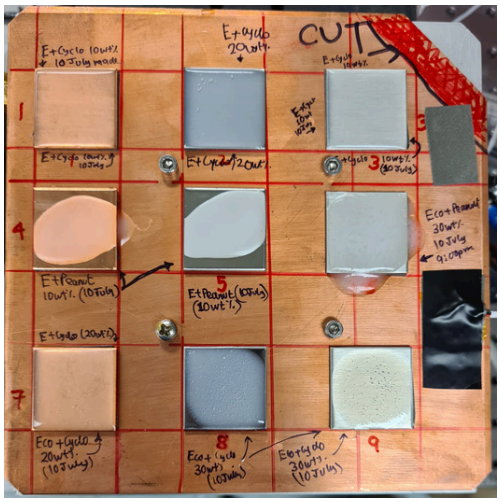
Experiment Setup



Picture from ongoing experiment

How?

- Worked in Prof. Sushant Anand's lab at the University of Illinois Chicago as part of a 5-member team - including Prof. Anand - under the direct guidance of PhD student Rukmava Chatterjee.
- Organized and executed lab tests to assess the adhesion strength of ice formed from various salt solutions on metallic surfaces.
- Performed polymer testing to determine suitability for cold-climate applications.
- Collected, analyzed, and visualized data using Excel, generating comparative reports for control vs experimental groups.
- Supported a study on PCMs that trap latent heat during condensation, delaying freezing.



Various metals and polymers



Tensile tests of polymers

Results

- Enhanced understanding of ice adhesion dynamics and PCM selection.
- Contributed to the development of a materials framework to reduce frost formation—relevant to energy systems, climate control, and medical applications.
- Recognized in a peer-reviewed journal, strengthening research credentials for future roles in materials, thermal, or robotics systems.



Team Photo

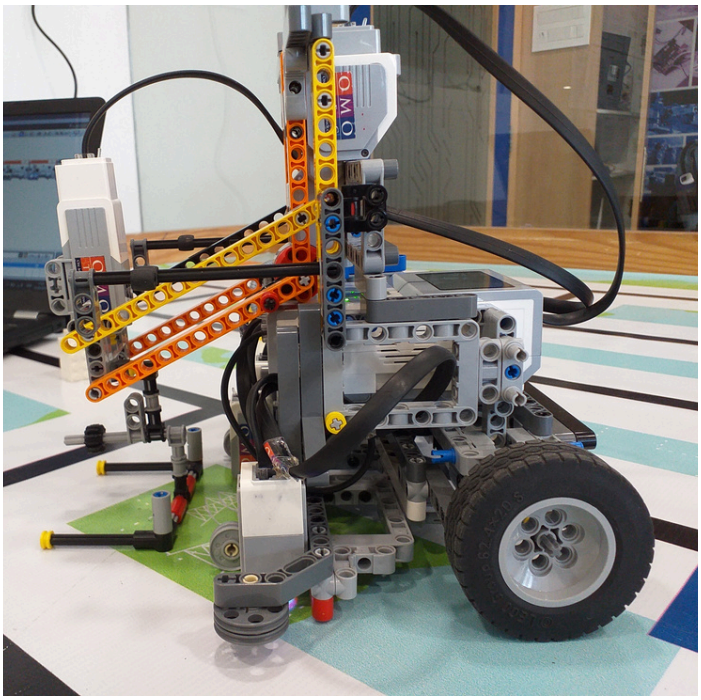
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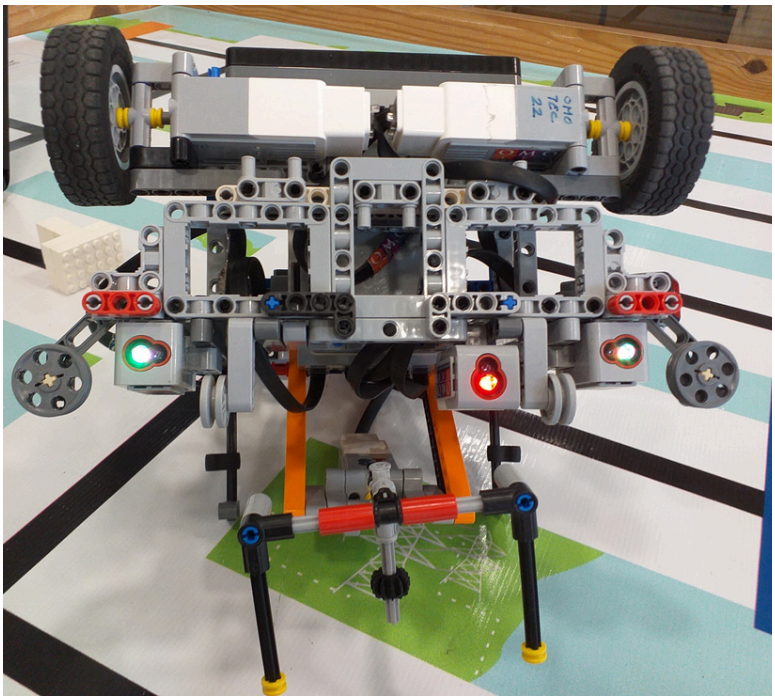
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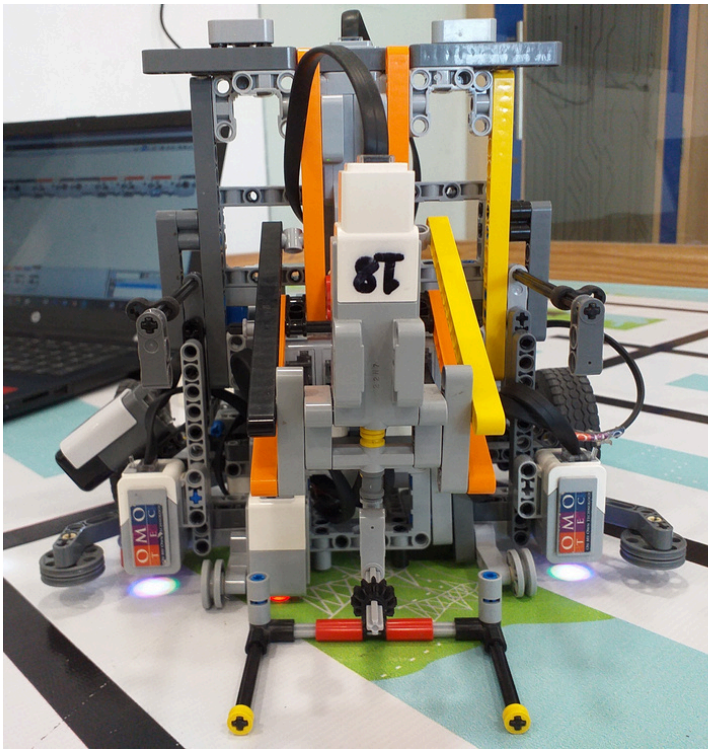
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WORLD ROBOT OLYMPIAD (WRO) SMART CITIES 2019







What?

- Participated in the World Robot Olympiad (WRO) 2019 Senior High category at the national level in India.
- Designed, built, and programmed autonomous robots to complete challenges related to the "Smart Cities" theme, focusing on tasks involving transportation and urban infrastructure.

How?

- Collaborated in a team of 2 members to design and build a robot using LEGO Mindstorms.
- Developed solutions through iterative design, testing, and refinement to optimize robot performance.
- Programmed the robot to autonomously navigate the course using black line following and perform tasks using sensors and pre-programmed instructions.
- Competed in multiple rounds, with robots judged on accuracy, speed, and reliability.
- Presented project details to judges, explaining design and programming choices.

Results

- My team secured 3rd position in the national competition.
- Our robot was successful in completing the maze autonomously with some minor mistakes.